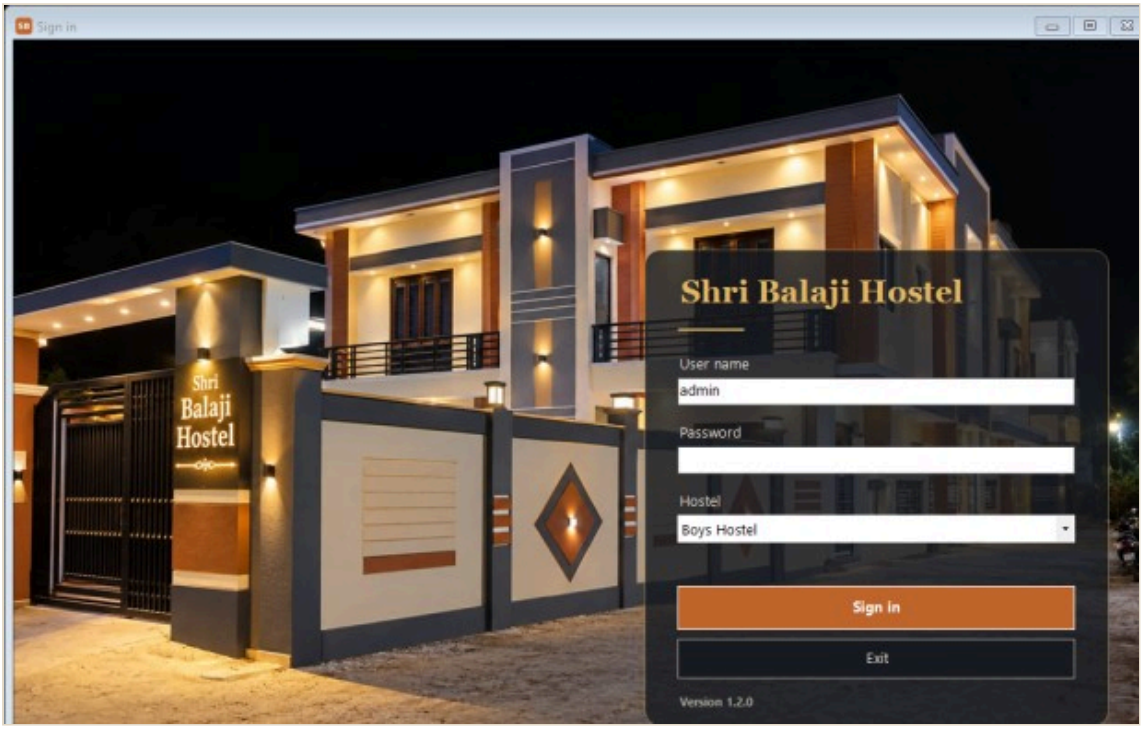
Shri Balaji Hostel

Hostel Management System: User Guide (version 1.5)



Contents

1. Getting started
2. Setting up a hostel
3. Students, father and mother, photos and I-cards
4. Rooms, check-in and the residency agreement
5. Night attendance
6. Fees and invoices
7. Payments and receipts
8. Fee reminders
9. Email (Gmail)
10. Dashboard and reports
11. Backup and restore
12. Admin account
13. Rules the program follows
14. When something goes wrong

1. Getting started

Start the program from the desktop shortcut **Hostel Management System**.

1. Enter the user name and password. A new installation uses **admin** / **admin**; change the password at once (section 12).
2. Choose the hostel to work on and click **Sign in**.

The menu on the left opens each screen. The hostel you work on is shown at the top right; change it there at any time. Colleges, rooms, students, invoices, payments, fee reminders, reports and the dashboard always show the **selected hostel only**.

> Fields marked with * are required. Messages in red under a form explain what to correct; nothing is saved until the form is correct.

2. Setting up a hostel

1. **Hostels**: click **Add** and enter the hostel name (address, phone and email are optional). Each new hostel gets the sharing types Single, Double and Triple.
2. **Colleges**: add the colleges whose students stay in the hostel.
3. **Rooms**: add the rooms: room number, floor, sharing type and whether the room is for boys or girls. The sharing type only decides **how many students fit in the room** (Single 1, Double 2, Triple 3). There is no rent per room: the fee is agreed with each student at check-in (section 4).

3. Students, father and mother, photos and I-cards

Add a student

1. **Students** > **Add**.

2. **Student** tab: only the **name** and the **gender** are required (the gender decides the rooms the student can be given). Everything else can be filled in later. A mobile number, email or Aadhaar number that is entered is checked for typing mistakes.
3. **College & Stay** tab: college (or "(not entered)"), course, class, admission date (today unless changed) and status.
4. **Father & Mother** tab: all optional: the father's name, mobile and email, and the mother's name, mobile and email.
   - Invoices, receipts and fee reminders go to the **father's email** (the mother's when the father has none).
   - Attendance emails go to the **mother's email** (the father's when the mother has none).
5. **Photo & Aadhaar** tab: click **Take Photo** to use the computer's webcam (the picture is live; click **Capture**, then **Use Photo**, or **Retake**), or **Choose Photo** to pick a JPG or PNG file. Add the Aadhaar card (image or PDF). Each file can be at most 5 MB.

Use the search box (student, father or mother name or mobile) and the filters (college, course, class, Active/Left) to find students. The Aadhaar number is shown masked (XXXX XXXX 1234) in lists and reports.

> If Take Photo says Windows does not allow the camera: open Windows **Settings** > **Privacy & security** > **Camera** and switch on **Camera access** and **Let desktop apps access your camera**.

I-cards

Select one or more students (hold **Ctrl** or **Shift** to select several) and click:

- **I-Card**: the student identity card with photo, name, father, college, room and bed, mobile numbers and **Valid till**.
- **Transport I-Card**: the transport pass with photo, only for students whose fee for this academic year includes transport.

Before the cards are made, the program asks for the **Valid till** date to print on them. It starts as the last day of the academic year; change it as needed (it cannot be in the past). The signature line on both cards reads **Hostel Management**.

The cards are wallet size (85.6 × 54 mm), eight on an A4 page. Choose where to save the PDF; it opens for printing. Students without a photo get an empty photo box, and the program tells you which ones.

4. Rooms, check-in and the residency agreement

Check-in and the student's fee

1. **Room Allocation** > **Check-in**.
2. Choose the student, the room (only rooms for the student's gender with a free bed are listed) and the **bed**. The lowest free bed is chosen for you; change it if needed.
3. Enter the **room rent for the year** and the **transport for the year** agreed with this student (leave transport empty if the student does not use it). Students in the same room can have different amounts. The total is shown below.

4. Click **Check-in**. The program creates the student's **invoice for the academic year** of the check-in date (July to June unless changed in Application Settings) with the room rent and transport. The full amount is due even when the student joins during the year.
5. After check-in the program offers to create the **residency agreement** PDF straight away.

Transfer and check-out

Select the student in **Room Allocation** and click **Transfer** (to another room, same day) or **Check-out**. Check-out sets the student to **Left** (inactive); the history of every room is kept. A transfer does not change the fee.

A student can check out **with or without paying**. When a fee is still pending, the check-out window shows the amount and asks:

- **Keep it pending**: the student still owes it; it stays on Fee Reminders and the student's invoice.
- **Waive it**: the rest of the fee is cancelled. The invoice then shows the fee, the amount waived at check-out and nothing pending.

Inactive students

Students who have checked out are on the **Inactive students** tab of the **Students** screen (current students are on **Active students**). For an inactive student you can:

- **View / Edit** the details and download the **Agreement PDF**.
- Give a **Refund** (see section 7).
- **Make Active**: for a student who comes back. Then allocate a room in **Room Allocation**.
- **Delete Permanently**: removes the student and **every record**: invoices, payments, refunds, room history, attendance, messages, photo, Aadhaar card and signatures. Type the reason. The program first makes a backup of the database (shown when done), so a mistake can be undone with Restore. Use it only when the records are really no longer needed.

Residency agreement

Students > select the student > **Agreement PDF**. Choose where to save it; it opens for printing. The agreement fills in the student's name, S/o or D/o with the father's name, address, Boys or Girls hostel, room and bed number, the number of beds in the room, the check-in date and the annual fee (the room rent plus transport agreed for the year, in figures and words). The student's **photo** is printed at the top right of the first page; without a photo the box says "Paste photo". Details that were not entered (address, mobile, father's name, college) print as a line to fill in by hand.

Signing the agreement on screen

Students > select the student > **Sign Agreement**. Four boxes open: the **student**, the **parent / guardian**, the **hostel owner** and a **witness** (type the witness's name too). Each person signs in their box with the mouse, the touchpad, a pen or a touch screen; **Clear** starts a box again.

- **Save and Make PDF** saves the signatures and creates the agreement with each signature above its line, the witness's name and "Signed on" with the date. **Save** only keeps the signatures; **Agreement PDF** prints them any time later.
- The owner signs once: with **Use for every agreement of this hostel** ticked, the owner's signature is filled in on every agreement of the hostel.
- Signatures are kept in **C:\HostelManagement\Documents\Signatures** and are part of every backup. Open Sign Agreement again to change or clear them.

To change the wording, click **Agreement Text** on the Students screen. Words in curly brackets, such as `{StudentName}` or `{BedNumber}`, are filled in for each student. **Restore Default** brings back the original text.

5. Night attendance

Attendance (under Students) lists the students who are in a room of the hostel that night, with room and bed. Everyone starts as **Present**.

1. At night, open **Attendance**; today's date is shown.
2. Untick **Present** for each student who is not in the hostel (add a remark if useful) and click **Save Attendance**.
3. The saved absent students appear on the right. Click **Email Parents** to email the **mother** of each absent student (the father when the mother has no email), with a copy to the admin. Each parent is emailed once per date; the program asks before emailing again.
4. Or click **WhatsApp Parents** to send the WhatsApp attendance message to the **mother's mobile** (the father's when the mother has none). The program shows an example and asks before sending; each parent gets it once per date (section 9a).

Choose another date to see or correct an earlier night. The text of the email is on **Settings** > **Email Settings** > **Attendance email**.

6. Fees and invoices

Each student has **one invoice per academic year** (July to June unless changed in Application Settings) with the room rent and the transport agreed for that year. It is created at check-in. Invoice numbers look like **SBH/2026-2027/0001**, and invoices and emails show the academic year as **2026-2027**. Invoices made before version 1.3 keep their number (for example SBH/2026-27/0003); new numbers carry on from there.

- **Edit Fee**: change the room rent or transport of the selected invoice (for example when a student stops using transport). The new total cannot be less than what has already been paid.
- **New Year Fees**: at the start of a new academic year, choose the year. The list shows every student still in a room who has no fee for that year, with last year's amounts for reference. Type each student's new room rent and transport and click **Create Fees**; students left empty are skipped.
- **View / Print** opens the invoice PDF: room rent, transport, total fee, **every payment received** (date, receipt, method, reference, amount), total paid and the amount pending. **Save PDF** saves it elsewhere, and **Email** sends it to the father (or the mother).
- Filters: search, status (Unpaid, Partly paid, Paid) and academic year. An invoice can be deleted only while it has no payments and no emails.

Example: room rent 50,000 + transport 12,000 = total 62,000. The student pays 10,000 at admission and 5,000 after 15 days: paid 15,000, pending 47,000.

7. Payments and receipts

1. **Invoices** > select the invoice > **Record Payment** (or **Payments** > **Record**, or **Fee Reminders** > **Record Payment**).

2. The pending amount is filled in; change it to the amount actually paid. The student can pay **any amount, any number of times**, but never more than the pending amount.
3. Choose Cash, UPI, Bank transfer or Cheque. For everything except cash, enter the UPI transaction id, bank reference or cheque number.

Each payment gets a receipt number such as **SBH/R/2026-2027/0001**. On the **Payments** screen, **View / Print** opens the receipt (with the amount in words and the balance still pending) and **Email Receipt** sends it to the father (or the mother) with the receipt PDF attached. Payment remarks are for the hostel only and are not printed on the receipt.

Correcting a wrong entry

- **Edit**: select the payment and correct the date, amount, method, reference or remarks, and type the reason (for example "wrong amount typed"). The receipt number stays the same; the receipt PDF shows the corrected details. The amount can never be more than the fee less the other payments.
- **Delete**: for a payment entered by mistake (for example twice). Type the reason; the amount becomes pending again.
- **Remarks**: every payment has a **View** button in the last column. It opens the full remarks of that payment in a popup. To change a remark, use **Edit**.
- **Change History**: every edit and deletion (also deleted refunds) with the date and time, receipt, student, what changed (old and new values) and the reason. Deletions are shown in red.

Refunds

Money paid back to a student, for example when the student leaves early. A refund can be **full or partial**, given **any number of times**, and also **after the student has checked out**.

1. **Payments** > select a payment of the student > **Refund**, or **Students** (either tab) > select the student > **Refund**.
2. Choose the fee (invoice) to refund from. The window shows what was paid and how much can still be refunded.
3. Enter the date, the amount, how it was paid back (Cash, UPI, Bank transfer or Cheque, with the reference except for cash) and the reason.
4. After saving, the program offers to open the **refund receipt** (number such as **SBH/RF/2026-2027/0001**, with the amount in words) and to email it to the father (or the mother).

A refund lowers both what was paid and the fee, so the pending amount does not change. The refund can never be more than what was paid on that invoice after earlier refunds. **Payments** > **Refunds** lists every refund with **View / Print**, **Email Receipt** and **Delete** (for a refund entered by mistake; type the reason). A payment cannot be deleted or reduced below the refunds already given on its invoice. The invoice PDF lists the refunds, and the dashboard shows money received after refunds.

8. Fee reminders

Fee Reminders (under Billing) lists every student who still owes money, including students who have left, with the total fee, paid, pending, who the email goes to and when the last reminder was sent. There are no due dates: send a reminder whenever you want. Reminders are **never sent automatically**.

1. Tick the **Send** box of each student to remind (**Select All** ticks everyone who has an email; **Clear** unticks all).
2. On the right is the ready-made reminder text. Change it if you like; words in curly brackets such as `{Pending}` are filled in for each student. **Preview** shows exactly what the selected student's parent will receive.
3. Click **Send Reminder**. Each email goes to the father (or the mother when the father has no email), with the invoice PDF and a copy to the admin. Students without any parent email are listed afterwards.

Send WhatsApp sends the WhatsApp fee reminder to the ticked students' parents instead: to the **father's mobile** (the mother's when the father has none), with the total, paid and pending amounts and how to pay by UPI QR code (no bank account number is sent). The program shows an example and asks before sending; students without a valid mobile number are listed afterwards. **Last reminder** counts both emails and WhatsApp messages (section 9a).

Save as Default keeps a changed text for next time; **Saved Text** brings back the saved one. **Export PDF** saves the list of pending fees.

9. Email (Gmail)

All emails are sent from the hostel's Gmail account **shribalajihostelsuddhowala@gmail.com**, and a copy goes to the admin's email (section 12). Gmail needs a 16 letter **app password**, not the normal password.

Create the app password (once)

1. In a browser, sign in to the hostel's Gmail account and open **myaccount.google.com** > **Security**.
2. Turn on **2-Step Verification** (Google sends a code to the mobile number).
3. Open **myaccount.google.com/apppasswords**, type a name such as *Hostel App* and click **Create**.
4. Copy the 16 letters shown in the yellow box (Google shows them only once).

Enter it in the program

1. **Settings** > **Email Settings** > **Gmail account** tab: paste the app password and click **Save**.
2. Type an address in **Send test to** and click **Send Test Email**.

The tabs **Invoice email**, **Receipt email**, **Fee reminder email** and **Attendance email** hold the texts of the emails; words in curly brackets are filled in. **Email history** lists every email, sent or failed, with the reason for a failure.

> If the Gmail password is ever changed, Google cancels the app password: create a new one and enter it again. The app password is also needed again after moving to another PC.

9a. WhatsApp (fee reminders and attendance)

There are two ways to send, chosen on **Settings** > **Email Settings** > **WhatsApp**:

- **WhatsApp on this PC** (normal choice): the program opens each parent's chat in WhatsApp on this PC with the message already typed, and you press Send. It uses the hostel's own WhatsApp; no Meta account, approval or charge.

- **Meta's WhatsApp API**: messages are sent automatically. This needs an approved WhatsApp Business Platform account (below).

WhatsApp on this PC

1. Install **WhatsApp** (or WhatsApp Business) for Windows on this PC and link it to the hostel's phone (scan the QR code on the PC with the phone). Or use **WhatsApp Web** in the browser.
2. On the **WhatsApp** tab choose *WhatsApp on this PC*, choose where chats open (the WhatsApp app or WhatsApp Web), check the two messages and click **Save WhatsApp Settings**. Words in curly brackets such as `{Pending}` are filled in for each parent; **Restore Default Texts** brings back the original messages. Optionally choose the payment QR image.
3. On **Fee Reminders** tick the students and click **Send WhatsApp** (or on **Attendance** click **WhatsApp Parents**). A window shows the first parent's message:
   - **Open in WhatsApp** opens that parent's chat with the message typed. Check it and press **Send** in WhatsApp. For fee reminders the payment QR image is copied at the same time: press **Ctrl+V** and **Send** again in the chat to add the QR below the message. (**Copy QR Image** copies it again if needed.) Set or change the QR in **Email Settings** > WhatsApp > **Choose QR...**.
   - Come back and click **Sent - Next**; the message is recorded in Email history and the next parent is shown. **Skip** goes to the next parent without recording.

> Send only to parents who gave their number for hostel messages, and only what they expect (fees, attendance). Many unexpected messages from one number can get it limited by WhatsApp.

Meta's WhatsApp API

WhatsApp messages are sent from the hostel's WhatsApp number through Meta's WhatsApp Business Platform, as two message **templates** that Meta approves once: `fee_reminder` and `attendance_alert` (category Utility, English). Meta charges for each delivered message, roughly ₹0.11 to ₹0.12 each; nothing is sent automatically.

Set up on Meta (once)

1. On **developers.facebook.com**: register, **Create App** with the use case *Connect with customers through WhatsApp* and the hostel's business portfolio, then **Start using the API**.
2. Connect the hostel's number (on the WhatsApp Business app this is done with a Meta partner, so the app keeps working on the phone). Note the **Phone Number ID** and the **WhatsApp Business Account ID** on the API Setup page.
3. In **WhatsApp Manager**: approve the display name, add a payment method and create the two templates with the agreed text.
4. In **Business Settings** > **System users**: create a system user, assign the app and the WhatsApp account, and **Generate token** with *whatsapp_business_messaging*. Keep the token safe; never send it by chat or email.

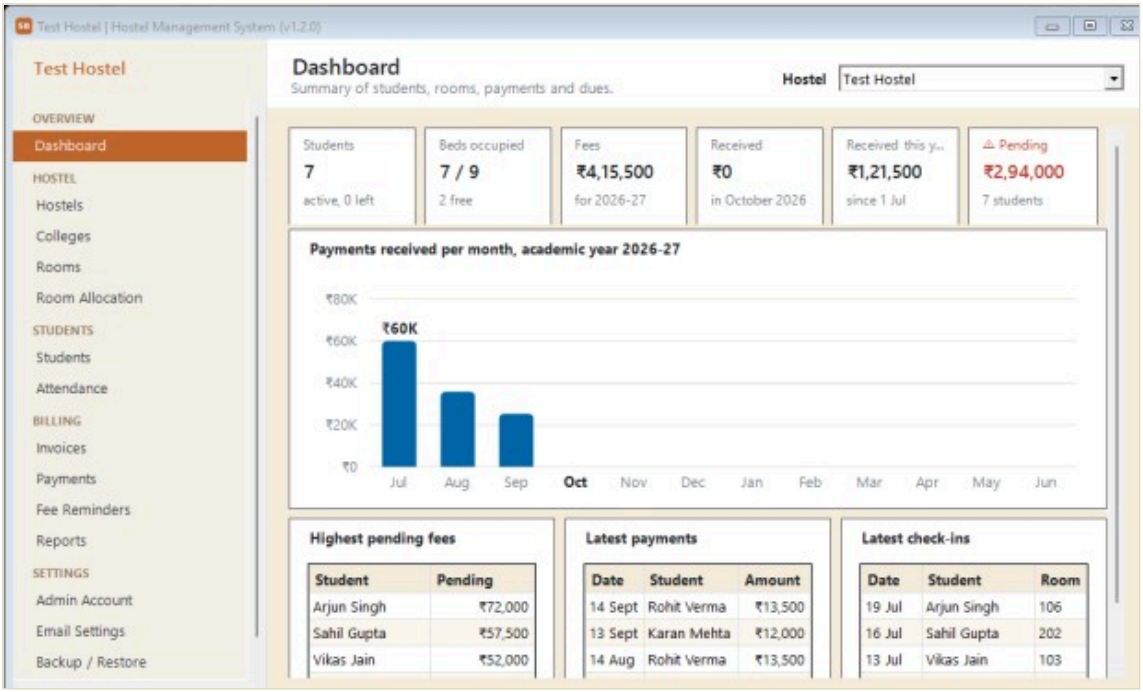
Enter it in the program

1. **Settings** > **Email Settings** > **WhatsApp** tab: choose *Meta's WhatsApp API*, then Phone Number ID, WhatsApp Business Account ID, the access token, the two template names (`fee_reminder`, `attendance_alert`) and the language `en`.

2. If the fee reminder template has the QR code as its header image, click **Choose QR...** and pick the payment QR image (PNG or JPG). Without an image header, leave it as *None*.
3. Click **Save WhatsApp Settings**, type a mobile number in **Send test to mobile** and click **Send Test Message**: Meta's "hello world" message should arrive.

The token is stored encrypted for this Windows user and never shown again (leave the box empty to keep it); after moving to another PC, enter it again. Sent and failed WhatsApp messages are listed in **Email history** with the mobile number and, for a failure, the reason (for example "not on WhatsApp").

10. Dashboard and reports

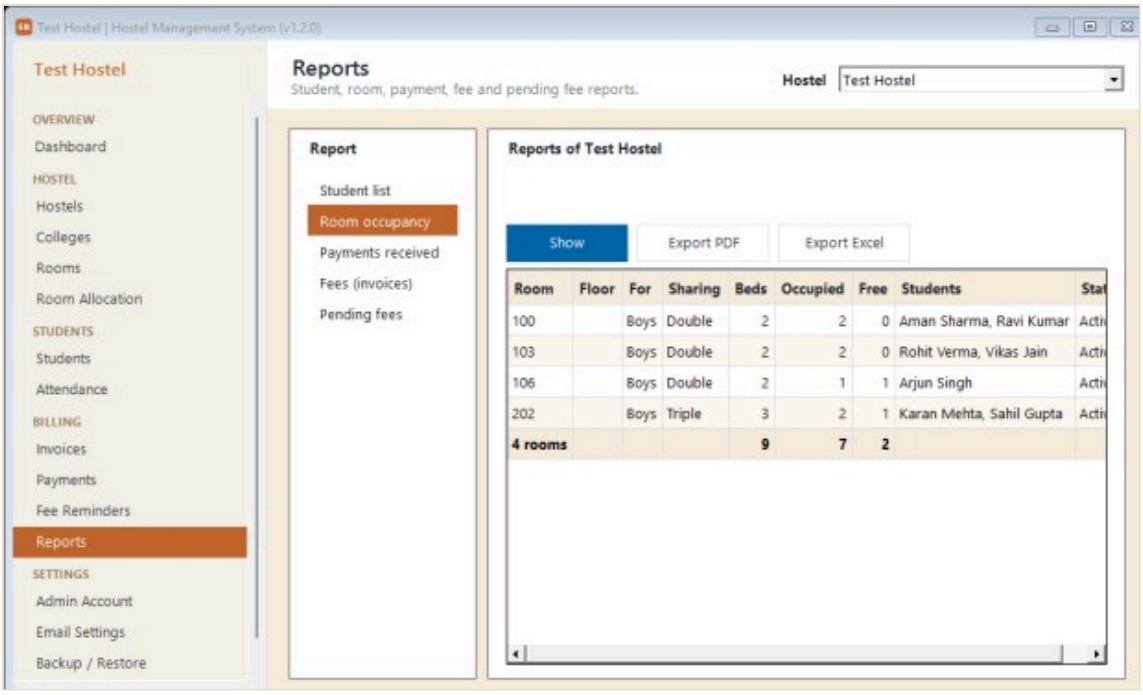


Dashboard with sample data

The **Dashboard** shows students, beds, the fees of this academic year, the amount received this month and this year, the pending fees, a chart of payments per month, and short lists of the students with the highest pending fees, the latest payments and check-ins.

Reports: choose a report on the left, set the filters, click **Show**, then **Export PDF** or **Export Excel**.

Report	Filters	Shows
Student list	Active/Left, college	Students with course, room, mobile, masked Aadhaar and father
Room occupancy	none	Each room with beds, occupied, free and students
Payments received	dates, payment method	Payments with a total per day and per method
Fees (invoices)	academic year, status	Each student's room rent, transport, total, paid and pending
Pending fees	none	Students with an amount pending and their invoices



Reports: room occupancy

11. Backup and restore

A backup is one zip file with the database and all photos and Aadhaar cards, saved in **C:\HostelManagement\Backups**.

- An **automatic backup** is made once a day when the program is closed; the last 30 are kept.
- **Backup Now** makes a backup at any time; these are never deleted.
- **Second copy in**: choose a USB drive or a Google Drive / OneDrive folder; every backup is copied there too, so it survives if the PC fails.

Restore: select a backup and click **Restore Selected...** (or **Restore From File...**). All current data is replaced by the backup; the program first saves a backup of the current data (so a restore can be undone) and then restarts.

12. Admin account

Settings > **Admin Account**: enter the admin's email (it receives a copy of every email sent to a father or mother) and phone, and change the password (at least 6 characters; enter the current password first).

Academic year

Settings > **Application Settings** > **Academic year**: choose the month the year **starts on** (always the 1st of that month; 1 July unless changed). The **end date** follows by itself: the day before the next year starts (30 June for a July start, 31 March for an April start). The screen also shows the current year with its dates. Click **Save**.

Fees, invoice numbers, New Year Fees, the dashboard, reports and I-cards all follow this year. Fees already made keep their academic year, so choose the start month once, before the first admissions, and change it only when the hostel's session really changes.

13. Rules the program follows

Topic	Rule
Academic year	Starts on the 1st of the month chosen in Application Settings (1 July unless changed) and ends the day before the next one starts; shown as 2026-2027 (or 2026 for a January start).
Fee	Agreed per student at check-in: room rent and transport for the year. One invoice per student per academic year; the full amount is due even when joining mid-year. A new fee every year (New Year Fees).
Rooms	Boys or girls; the sharing type decides the number of beds only; one student per bed. Transfer and check-out on any day. Check-out is allowed with a fee pending: keep it pending or waive it.
Refunds	Full or partial, any number of times, also after check-out; never more than what was paid. Pending does not change. Every refund has a receipt.
Students	Checked-out students are inactive (separate tab). They can be made active again or deleted permanently with all their records, after a backup.
Payments	Any amount, any number of times, never more than pending. A wrong entry can be edited or deleted; every change is kept in the Change History. Balance = total fee minus everything paid. Reference required except for cash. No GST, deposit, discounts or late fee.
Reminders	No due dates; the admin sends fee reminders whenever he wants, never automatically.
Emails	Sent only when you click a button. Fees: father, else mother. Attendance: mother, else father. Copy to the admin.
WhatsApp	Sent only when you click a button, as Meta-approved templates. Fee reminder: father's mobile, else mother's. Attendance: mother's mobile, else father's; once per date unless you choose to send again.
I-cards	Valid till the last day of the academic year. Transport I-card only with transport in this year's fee.

14. When something goes wrong

Message	What to do
The Microsoft Access Database Engine was not found	Install "Microsoft Access Database Engine 2016 Redistributable" (64 bit), see Installation.txt in the program folder.
The database could not be opened	Close Microsoft Access or any other program that has the database open and try again.
Gmail did not accept the sender address and app password	Create a new app password (section 9) and enter it in Email Settings.
Could not connect to smtp.gmail.com	Check the internet connection and try again.
A file is in use / could not be saved	Close the PDF or Excel file that is open and try again.
Windows does not allow this program to use the camera	Windows Settings > Privacy & security > Camera: switch on Camera access and Let desktop apps access your camera.
No camera was found	Connect a USB webcam or switch on the laptop camera, or use Choose Photo.
Anything else	The message says what happened; technical details are written to C:\HostelManagement\Logs. Send that file with your question.